Denote V be a Vector space over a field F.

Definition 0.0.1. Let $S \subseteq V$ be a subset.

- Define linear-combinations of S as:

$$\{a_1v_1 + \cdots + a_nv_n \mid n \in \mathbb{N}, a_i \in F, v_i \in S\}$$

- A subset $S \subseteq V$ is said to be *linearly independent* if: for any finite subset $\{v_1, \ldots, v_n\} \subseteq S$,

$$\sum_{i=1}^{n} a_iv_i = 0 \implies a_1 = \cdots = a_n = 0.$$

Lemma 0.0.2. Let $S \subseteq V$ be a subset. Then,

$$\langle S \rangle = \{a_1v_1 + \cdots + a_nv_n \mid n \in \mathbb{N}, a_i \in F, v_i \in S\}.$$

Proof. Denote the right term as $\overline{S}$. Clearly, $S \subseteq \overline{S} \subseteq V$.
Since $\langle S \rangle = \bigcap_{S \subseteq W \leq V} W$ is the smallest subspace containing S, $\langle S \rangle \subseteq \overline{S}$.
Conversely, $\overline{S} \subseteq \langle S \rangle$ because $\langle S \rangle$ contains S and is closed under the operations. □

Lemma 0.0.3. Let $S \subseteq V$ be a subset. TFAE:

a) S is linearly independent.

b) If $v \in V$ is a linear combination of S, then this representation is unique.

Proof. a) $\implies$ b) If there exist two representations of v by linear combination of S:

$$v = a_1v_1 + \cdots + a_nv_n = b_1v_1 + \cdots + b_nv_n$$

This implies that

$$(a_1 - b_1)v_1 + \cdots + (a_n - b_n)v_n = 0.$$

That is, for all $1 \leq i \leq n$, $a_i - b_i = 0$.
b) $\implies$ a) Let finite vectors $v_1, \ldots, v_n \in S$ be given.
If $a_1v_1 + \cdots + a_nv_n = 0$ for some $a_i \in F$, then $a_1, \ldots, a_n$ must be zero, by uniqueness of representation of zero. □

Theorem 0.0.4. Let $\beta \subseteq V$ be a subset. TFAE:

a) β is a basis of V.

b) For any $x \in V$, there exists a unique representation of x by linear combination of β.

Theorem 0.0.5. If a finite subset $S \subseteq V$ generates V, then there exists a basis β of V in S.

Proof. If $S \subseteq \{0\}$, then $V = \langle S \rangle = \{0\}$. Thus, $\emptyset \subseteq S$ is a basis of V.
Suppose S contains a non-zero vector $u_1 \in S$. Then, for any $u_2 \in S \setminus \{u_1\}$,

$$a_1u_1 + a_2u_2 = 0 \implies a_1 = a_2 = 0$$

Because if either $a_1 \neq 0$ or $a_2 \neq 0$, the cancellation gives $u_2 \in \langle u_1 \rangle$, contradiction.
Thus $\{u_1, u_2\} \subseteq S$ is linearly independent. (If $S \setminus \{u_1\} = \emptyset$, then $\{u_1\}$ is a basis of V).
Similarly, for $u_3 \in S \setminus \langle \{u_1, u_2\} \rangle$, $\{u_1, u_2, u_3\} \subseteq S$ is linearly independent.
Since S is finite, this process will terminate in finite iterations.
That is, for some $n \in \mathbb{N}$, $S \subseteq \langle \{u_1, \ldots, u_n\} \rangle$. This $\{u_1, \ldots, u_n\}$ is a basis of V, by $V = \langle S \rangle \subseteq \langle \{u_1, \ldots, u_n\} \rangle$. □

Replacement Theorem

> **Theorem 0.0.6.** Suppose that $G \subseteq V$ generates V, $L \subseteq V$ is linearly independent, with $\#(G) = n, \#(L) = m$. Then, $m \leq n$ and there exists $H \subseteq G$ s.t. $\#(H) = n - m$ and $\langle L \cup H \rangle = V$.

Proof. Using Induction for m.
If $m = 0$ (that is, $L = \emptyset$), then take $H = G$. Now suppose for $m \in \mathbb{N}$, the statement holds.
Let $L = \{v_1, \dots, v_m, v_{m+1}\}$ be a linearly independent subset of V, and G be a generating subset of V with $\#(G) = n$.
Since a subset of a linearly independent set is linearly independent, $\{v_1, \dots, v_m\} \subseteq L$ is linearly independent.
Now, there exists a subset $\{u_1, \dots, u_{n-m}\} \subseteq G$ such that

$$\langle \underbrace{\{v_1, \dots, v_m\} \cup \{u_1, \dots, u_{n-m}\}}_{K} \rangle = V.$$

That is, there exists a linear combination of K for v_{m+1} as

$$\sum_{i=1}^{m} a_i v_i + \sum_{j=1}^{n-m} b_j u_j = v_{m+1}.$$

If $n - m = 0$, then $v_{m+1} = \sum_{i=1}^{m} a_i v_i$, which contradicts linearly independence.
Now, $n \geq m + 1$. And b_j are not all zero; if all zero, then it contradicts linearly independence.
For some $1 \leq j \leq n - m$, take $b_j \neq 0$. Then,

$$u_j = b_j^{-1} \cdot \left(v_{m+1} - \sum_{i \neq j} b_i u_i - \sum_{i=1}^{m} a_i v_i \right) \in \langle L \cup H \rangle$$

where $H = \{u_1, \dots, u_{j-1}, u_{j+1}, \dots, u_{n-m}\}$. Since $K \subseteq \langle L \cup H \rangle$ and $\langle K \rangle = V$, $\langle L \cup H \rangle = V$. $\square$

Corollary 0.0.7. If V has a finite basis, then every basis of V has the same cardinality.

> **Theorem 0.0.8.** Every Vector Space has a Basis.

Proof. Using Zorn's Lemma. Let V be a Vector Space. Put $P \overset{\text{def}}{=} \{S \subseteq V \mid S \text{ is linearly independent subset}\}$.
Clearly, $\{v\} \in P$ for any $v \in V$. Thus, $(P, \subseteq)$ is non-emepty partially ordered set which has inclusion as order.
Suppose $\mathcal{C} \subseteq P$ is a chain of P. If $\mathcal{C} = \emptyset$, then $\{v\}$ is upper bound of $\mathcal{C}$ for any $v \in V$. Thus, assume $\mathcal{C} \neq \emptyset$.
To show that $\mathcal{C}$ has an upper bound in P, denote the union of all elements in the chain:

$$B = \bigcup_{\beta \in \mathcal{C}} \beta$$

Clearly this is upper bound of $\mathcal{C}$. Hence, enough to show that B is linearly independent; That is, $B \in P$.
Suppose that B is linearly dependent. Then, there exist finite vectors in B and scalars such that

$$a_1 v_1 + \cdots + a_n v_n = 0$$

for some a_j is not zero. Since $\mathcal{C}$ is a Chain, there exists some subset of $\mathcal{C}$ which containing $\{v_1, \dots, v_n\}$.
Because for each v_i, there exists $\beta_i \in \mathcal{C}$ such that $v_i \in \beta_i$, and

$$\{v_1, \dots, v_n\} \subseteq \bigcup_{i=1}^{n} \beta_i \overset{(*)}{\in} \mathcal{C}$$

The inclusion $(*)$ is given by finitely construction: for any $\beta_i, \beta_j \in \mathcal{C}$, $\beta_i \subseteq \beta_j$ or $\beta_i \supseteq \beta_j$.
Thus, the union of two elements in the chain is either one. This is contradiction that the finite union is linearly independent, being it is contained in P. In sum, $\mathcal{C}$ has an upper bound in P.
By Zorn's Lemma, there exists a Maximal $\mathcal{M} \subsetneq V$ which is linearly independent, and satisfies

If $\beta \subseteq V$ is linearly independent, then $\beta \subseteq \mathcal{M}$.

Lastly, if $\langle \mathcal{M} \rangle \subsetneq V$, then there exists $v \in V \setminus \langle \mathcal{M} \rangle$ such that $\mathcal{M} \subsetneq \mathcal{M} \cup \{v\}$.
But, this implies $\mathcal{M} \cup \{v\}$ is linearly independent and proper supset of $\mathcal{M}$, contradiction.
Consquently, $\langle \mathcal{M} \rangle = V$. $\square$